



MINOR PROJECT · IBM SkillsBuild Internship PRESENTATION

PaveMind AI

Intelligent Road Pothole Detection & Severity Analysis Platform

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PROJECT DOMAIN

Artificial Intelligence & Machine Learning · Computer Vision

Group 74 · AIML74 · UPES Dehradun · IBM SkillsBuild Internship 2025–26



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01 Introduction

TECHNICAL BACKGROUND OF PROJECT

PaveMind AI is a computer-vision system that detects potholes in dashcam road footage, estimates their real-world size and severity, and reports results through a live analytics dashboard.

TECHNICAL CONCEPTS (ALGORITHMS) USED

YOLOv12 Nano object detection, camera-geometry perspective projection for physical sizing, and a multi-factor weighted severity-scoring algorithm.

MOTIVATION

Reactive, complaint-driven repair cycles leave road defects unaddressed for weeks and give municipal teams no quantified way to decide which road to fix first.

PROBLEM STATEMENT

Design and deploy an end-to-end system that detects potholes from video, measures their physical dimensions without 3D sensors, classifies severity, and ranks roads for maintenance priority.

AREA OF APPLICATION

- Municipal road-maintenance departments
- Fleet-mounted inspection (buses, utility trucks)
- Smart-city infrastructure monitoring
- Highway authority condition surveys

DATASET & INPUT FORMAT

Input: Dashcam image / video (.jpg, .mp4, .webm)

Sources: Kaggle Pothole Dataset + CRDDC / RDD2022 (D40)

Merged size: 765 images · 2,024 pothole instances

Coverage: Road conditions across India, USA, Japan



02 Literature Review

CITED RELATED WORK — FROM SENSORS TO REAL-TIME DETECTION

EARLY / 2015+

Vibration Sensors

Accelerometers detect road-induced jolts. Effective in controlled tests but highly sensitive to vehicle suspension and speed.

2015+

CNN Classifiers

ResNet / VGG sliding-window detectors improve accuracy over sensors but are too slow for real-time dashcam use.

2016+

YOLO Family

Redmon et al. unify detection and classification in one forward pass. YOLOv5→v12 steadily improve accuracy and efficiency.

2024

PaveMind AI

YOLOv12n + geometric sizing + severity scoring + full deployment — goes beyond bounding boxes to actionable output.

GAP IN LITERATURE

Published systems typically stop at bounding-box localisation. They rarely provide real-world physical dimensions, multi-tier severity classification, or an end-to-end deployed system.

INFERENCE FROM LITERATURE

The evolution from sensors to single-pass detectors shows real-time detection is achievable; PaveMind AI's contribution is combining detection with measurement, scoring, and deployment in one pipeline.



03 Objective

1 Main Objective

To design and deploy an AI-based system that detects potholes from road video, estimates their real-world physical size using camera geometry, classifies severity, and produces a single road-health score for maintenance prioritisation.

2 Sub Objectives — Five Research Contributions

1

Fine-tune YOLOv12 Nano (2.6 M params) achieving mAP@0.5 of 91.23% on a merged 765-image dataset.

2

Build a camera-geometry-based physical size estimator that needs no 3D sensor hardware.

3

Design a multi-factor severity score combining size, depth, confidence, aspect ratio and temporal persistence.

4

Develop a seven-stage false-positive mitigation pipeline against sky, clouds, tyre tracks and noise.

5

Deploy an end-to-end system — FastAPI + MongoDB + React 19 — with a live GPS heatmap dashboard.

04 Methodology

PIPELINE STEPS

- 1 Video / image input uploaded via the React dashboard
- 2 YOLOv12 Nano inference detects candidate potholes per frame
- 3 Seven-stage geometric + confidence gate rejects false positives
- 4 Camera-geometry projection converts pixels to real-world cm
- 5 Multi-factor scoring assigns Minor / Moderate / Severe class
- 6 Road Health Index (0–100) aggregated and pushed to MongoDB / dashboard

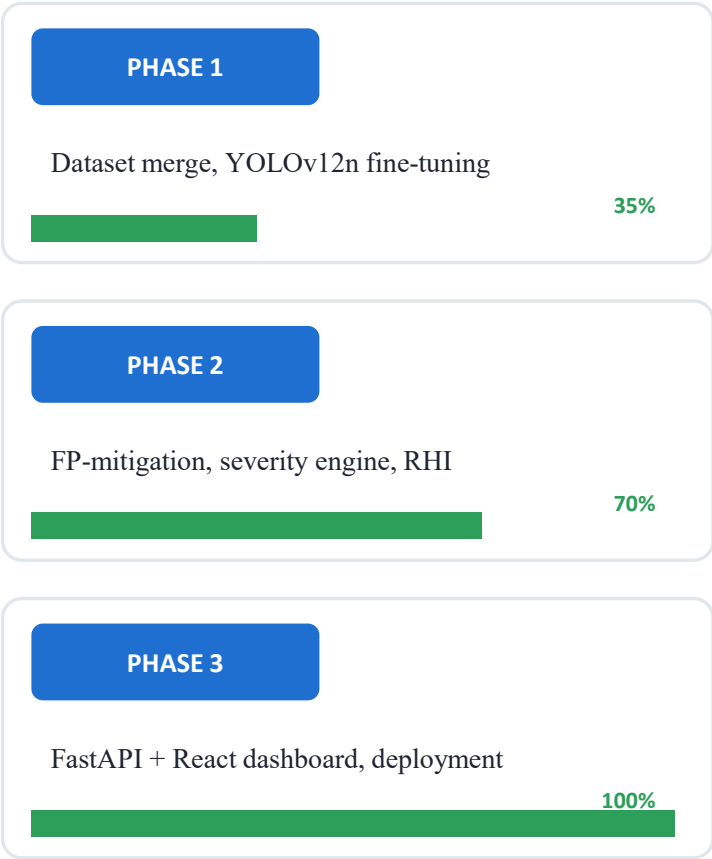
TECH STACK

AI / Vision: YOLOv12n · Ultralytics · ByteTrack · OpenCV · NumPy

Backend: FastAPI · MongoDB Atlas · PyMongo · imageio + FFmpeg

Frontend: React 19 + Vite · Leaflet · Recharts

DEVELOPMENT TIMELINE



05 Working Model

SYSTEM ARCHITECTURE — THREE LAYERS, ONE DEPLOYABLE SYSTEM

01

Core Analysis Engine

ML inference & analysis core — SeverityEngine, AlertSystem, AnalyticsManager, Visualizer built on Python, Ultralytics, OpenCV, NumPy.

02

REST API Backend

8 endpoints, non-blocking pipeline — FastAPI + MongoDB Atlas with transparent local-JSON fallback, ByteTrack tracking, H.264 export.

03

Web Dashboard

Premium React 19 + Vite SPA — KPI overview, drag-and-drop image analyzer, video tracker with live progress, Leaflet GPS heatmap.

SEVERITY CLASSES

MINOR < 0.45 **MODERATE** $0.45-0.70$ **SEVERE** ≥ 0.70

DELIVERABLES

- Web-based AI platform with automated pothole detection
- cm-accurate severity scoring and Road Health Index
- Interactive GPS heatmap and one-click CSV / JSON / media export



06 Output

ROAD HEALTH INDEX — THE HEADLINE METRIC

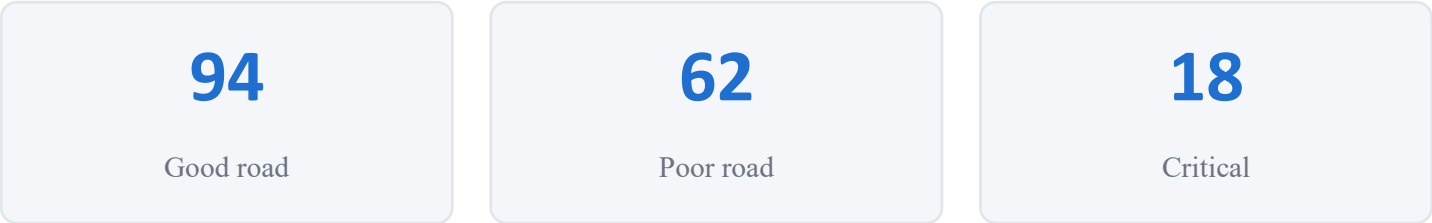
CORE CAPABILITIES

$$RHI = \max(0, 100 - 15 \times N_{\text{severe}} - 6 \times N_{\text{moderate}} - 2 \times N_{\text{minor}})$$

- 85–100 — Excellent
- 60–84 — Acceptable
- 30–59 — Poor
- 0–29 — Critical

- Annotated image analysis with severity labels
- Frame-by-frame video processing, re-encoded MP4
- Automatic Minor / Moderate / Severe grading
- Interactive GPS heatmap of detections
- One-click ZIP report: CSV + JSON + processed media

SAMPLE SCAN OUTPUT



07 Results

MODEL BENCHMARK — 25-EPOCH CONFIGURATION VS. 2-EPOCH BASELINE

91.23%

mAP@0.5

+56.6% vs 2-epoch

64.02%

mAP@0.5:0.95

+112.2% vs 2-epoch

88.04%

Precision

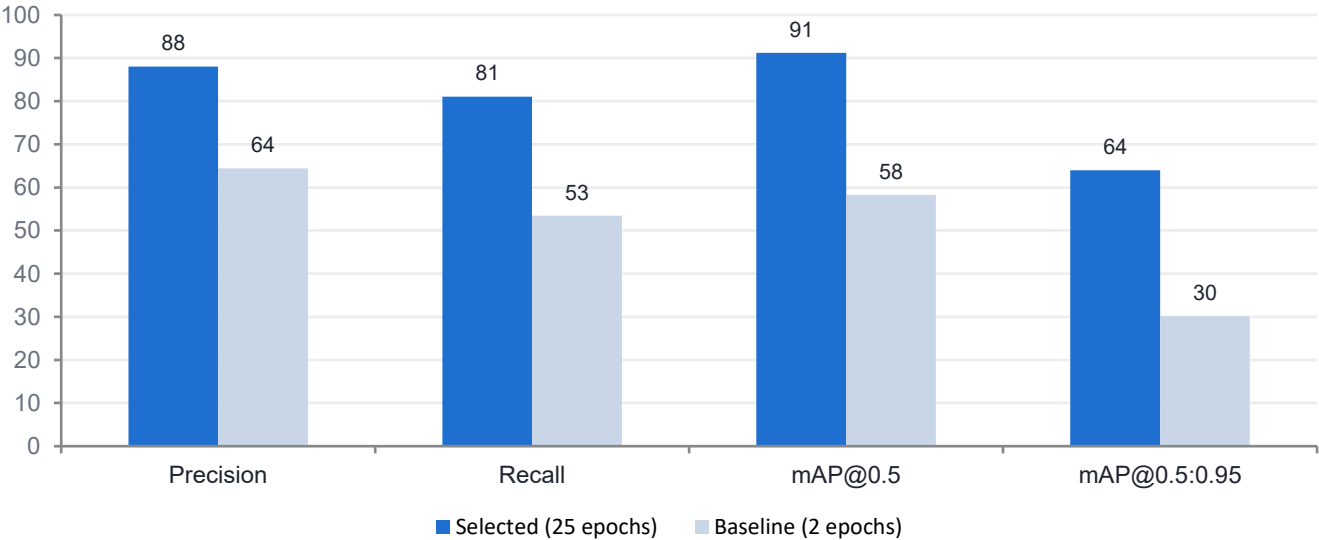
+36.6% vs 2-epoch

81.05%

Recall

+51.8% vs 2-epoch

Training Benchmark Comparison (%)



CPU RUNTIME

Avg CPU latency

281.4 ms / frame

Throughput

3.55 FPS (CPU)

Model size

2.56 M params · 6.3 GFLOPs

08 Comparative Study

LIVE DETECTION PERFORMANCE — REAL INDIAN ROAD FOOTAGE

692-frame sequence · 2,302 total detections · 5.5 FPS average on CPU · mean confidence 0.57 · avg. pothole width 73.7 cm.

692

Frames processed

2,302

Total detections

181.9 ms

Avg. per frame

5.5 FPS

on CPU

SEVERITY BREAKDOWN (692-FRAME TEST SEQUENCE)

SEVERE 852 (37.0%)

MODERATE 719 (31.2%)

MINOR 731 (31.8%)

COMPARATIVE STUDY — 25 VS. 2 TRAINING EPOCHS

Metric	25 Epochs	2 Epochs	Δ
Precision	88.04%	64.43%	+36.6%
Recall	81.05%	53.42%	+51.8%
mAP@0.5	91.23%	58.27%	+56.6%

COMPARATIVE STUDIES — MANUAL VS. AUTOMATED

- Manual road surveys are slow, subjective and inconsistent across inspectors; Pavemind AI produces faster, consistent, unbiased severity scores.
- Automatic ranking of roads removes the manual effort of prioritising repair order across hundreds of kilometres of road network.

09 Conclusion

JUSTIFICATION OF OBJECTIVES

The project successfully demonstrates an AI-based system for pothole detection, real-world size estimation, and severity scoring — achieving $mAP@0.5 = 91.23\%$ and Precision = 88.04% at just 2.6 M parameters, and delivering an end-to-end deployed FastAPI + MongoDB + React dashboard rather than a research notebook.

- Detection accuracy — 91.23% $mAP@0.5$ at 2.6M params
- Actionable output — real-world cm measurements, no 3D sensor
- Full deployment — REST API + MongoDB + React dashboard
- Robust in the wild — handles Indian, US, and Japanese road conditions

FUTURE SCOPE

- 1 **GPU Inference** — Deploy on CUDA server for 30+ FPS real-time video.
- 2 **Monocular Depth** — Integrate MiDaS to replace geometric depth approximation.
- 3 **Mobile Application** — React Native app for field inspection teams.
- 4 **Road Authority API** — Municipal integration with automated SMS / email alerts.
- 5 **Cloud Deployment** — Vercel + Hugging Face Spaces for public demo access.

10 References

PAPERS REFERENCED

- Redmon, J. et al. (2016) – You Only Look Once: Unified, Real-Time Object Detection.
- Ultralytics (2024) – YOLOv12: Attention-Centric Real-Time Object Detectors.
- Arya, D. et al. (2022) – RDD2022: A multi-national road damage dataset (CRDDC).
- Kaggle Pothole Detection Dataset – open road-surface imagery collection.

LIST OF CITED PAPERS

- Smart India Hackathon 2025 – Problem Statement, Ministry of Road Transport & Highways.
- Wang, C.-Y. et al. (2023) – Reversed Residual Networks and Efficient Layer Aggregation (ELAN).
- OpenCV / Ultralytics documentation – ByteTrack multi-object tracking.





Thank You

Questions & Discussion

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